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10.3.26-17.3.26

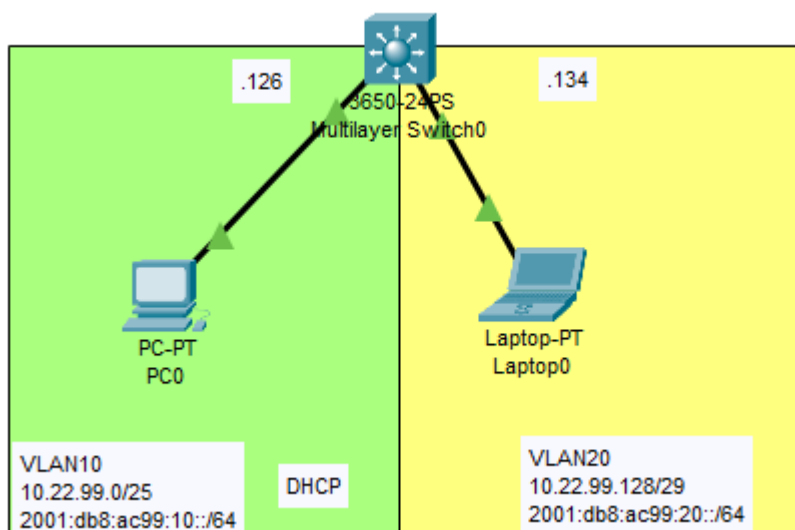


Arbeitsform:

Jede Schülerin und jeder Schüler muss aber einen **eigenständigen** Laborbericht abgeben. Duplikate bzw. weitreichende Kopien werden nicht akzeptiert. Router und Switchnamen sind zu personalisieren. Dokumentiert im Bericht am Deckblatt neben Namen und Datum auch Deine Katalognummer sowie den Namen Deiner Arbeitskolleg:innen.

Dauer:

4 Schulstunden.



Bitte heuer 10.24er Adressen verwenden.

Teil 1 - Szenario im Packettracer:

1. Dokumentiert und konfiguriert Euer Szenario auf Eurem persönlichen Laptop zuerst als Packettracer Szenario:

Ein Router Modell C3650 (SW C3750-IPSERVICESK9 im Labor) , sowie 2 PCs, passende Verkabelung zwischen Switch und PCs.

Verwendet das bereits bestehende Adressschema mit der Excel-Vorlage und **fügt dieses hier** ein.

VLAN	Gerät	IPv4-Adresse	Subnetzmaske	Gateway	Netzwerk
VLAN10	Switch (SVI)	10.26.3.1	255.255.255.128 (/25)	–	10.26.3.0
VLAN10	PC	10.26.3.10	255.255.255.128 (/25)	10.26.3.1	10.26.3.0
VLAN20	Switch (SVI)	10.26.3.129	255.255.255.128 (/25)	–	10.26.3.128

VLAN	Gerät	IPv4-Adresse	Subnetzmaske	Gateway	Netzwerk
VLAN20	Laptop	10.26.3.130	255.255.255.128 (/25)	10.26.3.129	10.26.3.128

2. L2-Konfiguration:

Konfiguriert am L3-Switch die VLANs und weist die physischen Interfaces den VLANs 10-user bzw. 20-admin zu.

```
Switch(config)#vlan 10
Switch(config-vlan)#name VLAN10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#interface gil/0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gil/0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#
Switch(config)#interface vlan 10
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan10, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, changed state to up

Switch(config-if)#ip address 10.26.3.1 255.255.255.128
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan20, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, changed state to up

Switch(config-if)#ip address 10.26.3.129 255.255.255.128
Switch(config-if)#no shutdown
Switch(config-if)#exit
```

3. Konfiguriert alle Ports mit dem Kommando:

spanning-tree portfast

Dadurch durchlaufe die Ports nach dem physical LinkUp keine („orange“) Lernphase (ob eine Schleife gesteckt wurde), sondern gehen sofort in den Forwarding Modus.

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range gil/0/1-2
Switch(config-if-range)#spanning-tree portfast
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/0/1 but will only
have effect when the interface is in a non-trunking mode.
%Warning: portfast should only be enabled on ports connected to a single
host. Connecting hubs, concentrators, switches, bridges, etc... to this
interface when portfast is enabled, can cause temporary bridging loops.
Use with CAUTION

%Portfast has been configured on GigabitEthernet1/0/2 but will only
have effect when the interface is in a non-trunking mode.
Switch(config-if-range)#exit
Switch(config)#
```

4. Aktiviert IPv4 Routing durch das Kommando

ip routing

```
Switch(config)#ip routing
```

5. Vergebt auf den Switch Virtual Interfaces (SVI) die IP-Adressen (Default-GWs) aus Eurem Schema.

Konfiguriert ebenso IPv6!

```
Switch(config)#  
Switch(config)#interface vlan 10  
Switch(config-if)#ipv6 address 2001:db8:26:10::1/64  
Switch(config-if)#no shutdown  
Switch(config-if)#exit  
Switch(config)#interface vlan 20  
Switch(config-if)#ipv6 address 2001:db8:26:20::1/64  
Switch(config-if)#no shutdown  
Switch(config-if)#exit  
Switch(config)#ipv6 unicast-routing  
Switch(config)#
```

6. DHCP-Konfiguration.

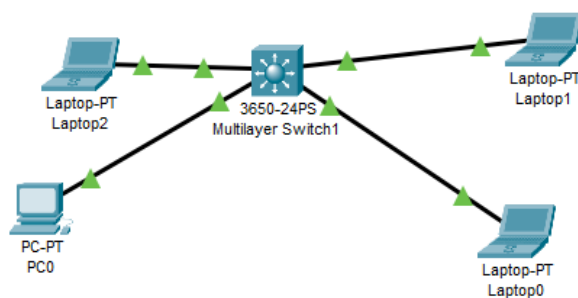
Konfiguriert am Router für jedes VLAN einen DHCP-Server.

Konfiguriert **jeweils** einen DHCP-Pool für jedes VLAN-Interface:

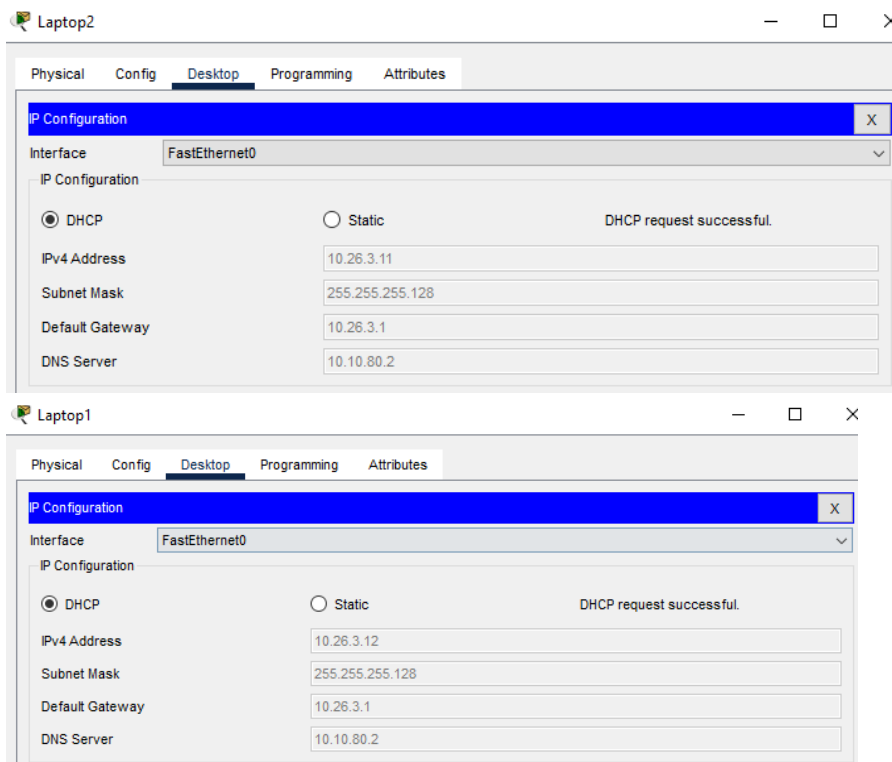
```
ip dhcp pool vl10  
network 10.23.99.0 255.255.255.128  
default-router 10.23.99.126  
dns-server 10.10.80.1  
dns-server 10.10.80.2  
  
Switch#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#ip dhcp excluded-address 10.26.3.1 10.26.3.10  
Switch(config)#ip dhcp excluded-address 10.26.3.129 10.26.3.135  
Switch(config)#ip dhcp pool vl10  
Switch(dhcp-config)#network 10.26.3.0 255.255.255.128  
Switch(dhcp-config)#default-router 10.26.3.1  
Switch(dhcp-config)#dns-server 10.10.80.1  
Switch(dhcp-config)#dns-server 10.10.80.2  
Switch(dhcp-config)#exit  
Switch(config)#ip dhcp pool vl20  
Switch(dhcp-config)#network 10.26.3.128 255.255.255.128  
Switch(dhcp-config)#default-router 10.26.3.129  
Switch(dhcp-config)#dns-server 10.10.80.1  
Switch(dhcp-config)#dns-server 10.10.80.2  
Switch(dhcp-config)#
```

7. Steckt PCs an die jeweiligen Switchports und kontrolliert (Nachweis mit screenshots), dass

```
Switch(config)#interface gil/0/3  
Switch(config-if)#switchport mode access  
Switch(config-if)#switchport access vlan 10  
Switch(config-if)#no shutdown  
Switch(config-if)#exit  
Switch(config)#interface gil/0/4  
Switch(config-if)#switchport mode access  
Switch(config-if)#switchport access vlan 10  
Switch(config-if)#no shutdown  
Switch(config-if)#exit  
Switch(config)#
```



- a. Die PCs via DHCP IP Adressen aus dem passenden Pool bekommen.



- b. dokumentiert ein DHCP-Discover-Paket, DHCP-Offer, -Request und Acknowledge-Paket im Simulation-Mode (Packet Details) und kläre ob es sich um Broadcast/Multicast oder Unicast-Frames handelt!

PDU Information at Device: Multilayer Switch1

OSI Model Inbound PDU Details Outbound PDU Details

At Device: Multilayer Switch1
Source: Laptop2
Destination: 255.255.255.255

In Layers	Out Layers
Layer 7: DHCP Packet Server: 0.0.0.0, Client: 0.0.0.0	Layer 7: DHCP Packet Server: 0.0.0.0, Client: 0.0.0.0
Layer 6	Layer 6
Layer 5	Layer 5
Layer 4: UDP Src Port: 68, Dst Port: 67	Layer 4
Layer 3: IP Header Src. IP: 0.0.0.0, Dest. IP: 255.255.255.255	Layer 3
Layer 2: Ethernet II Header 0004.9A50.3114 >> FFFF.FFFF.FFFF	Layer 2: Ethernet II Header 0004.9A50.3114 >> FFFF.FFFF.FFFF
Layer 1: Port GigabitEthernet1/0/3	Layer 1: Port(s):

1. GigabitEthernet1/0/3 receives the frame.

PDU Information at Device: Multilayer Switch1

OSI Model Inbound PDU Details Outbound PDU Details

PDU Formats

EthernetII

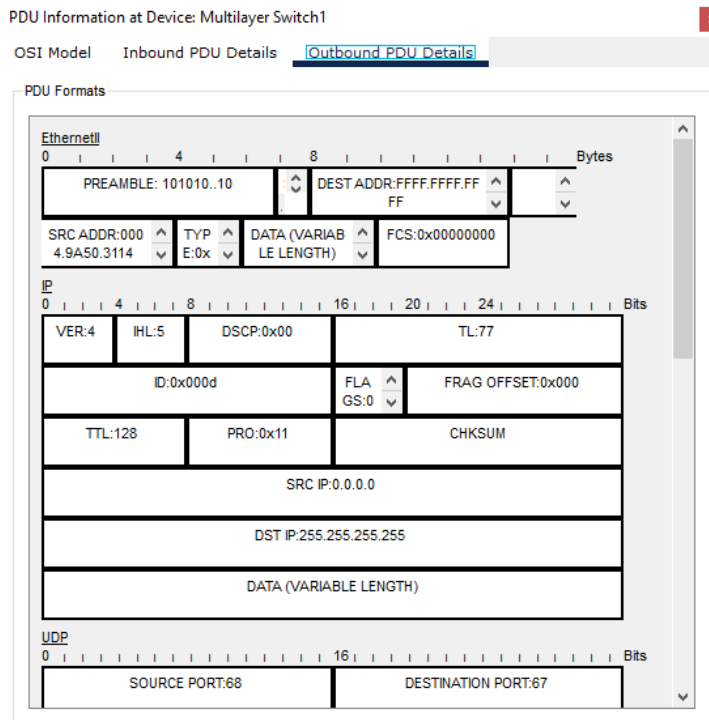
0		4		8		16		20		24		Bytes	
PREAMBLE: 101010...10						DEST ADDR: FFFF.FFFF.FF						FF	
SRC ADDR: 0004.9A50.3114				TYP: 0x0		DATA (VARIABLE LENGTH)				FCS: 0x00000000			

IP

0		4		8		16		20		24		Bits	
VER: 4		IHL: 5		DSCP: 0x00		TL: 77							
ID: 0x000d						FLAG: 0		FRAG OFFSET: 0x000					
TTL: 128				PRO: 0x11				CHKSUM					
SRC IP: 0.0.0.0													
DST IP: 255.255.255.255													
DATA (VARIABLE LENGTH)													

UDP

0		4		8		16		20		24		Bits	
SOURCE PORT: 68						DESTINATION PORT: 67							



- c. Die PCs das Default-GW und eine Routeradresse hinter dem DG erreichen können.

PC0

Physical Config Desktop Programming Attributes

Command Prompt

```

Reply from 10.26.3.1: bytes=32 time<1ms TTL=255
Reply from 10.26.3.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.26.3.1:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

Control-C
^C
C:\>ping 10.26.3.129

Pinging 10.26.3.129 with 32 bytes of data:

Reply from 10.26.3.129: bytes=32 time<1ms TTL=255
Reply from 10.26.3.129: bytes=32 time<1ms TTL=255

Ping statistics for 10.26.3.129:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

```

- d. Ein PC in einem anderen Subnet erreicht werden kann.

```

C:\>ping 10.26.3.130

Pinging 10.26.3.130 with 32 bytes of data:

Reply from 10.26.3.130: bytes=32 time<1ms TTL=127
Reply from 10.26.3.130: bytes=32 time<1ms TTL=127

Ping statistics for 10.26.3.130:
    Packets: Sent = 2, Received = 2, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

Control-C
^C
C:\>

```

- e. gebt am Router die Belegung der DHCP-Pools mit
show ip dhcp pool vl20
aus.

```

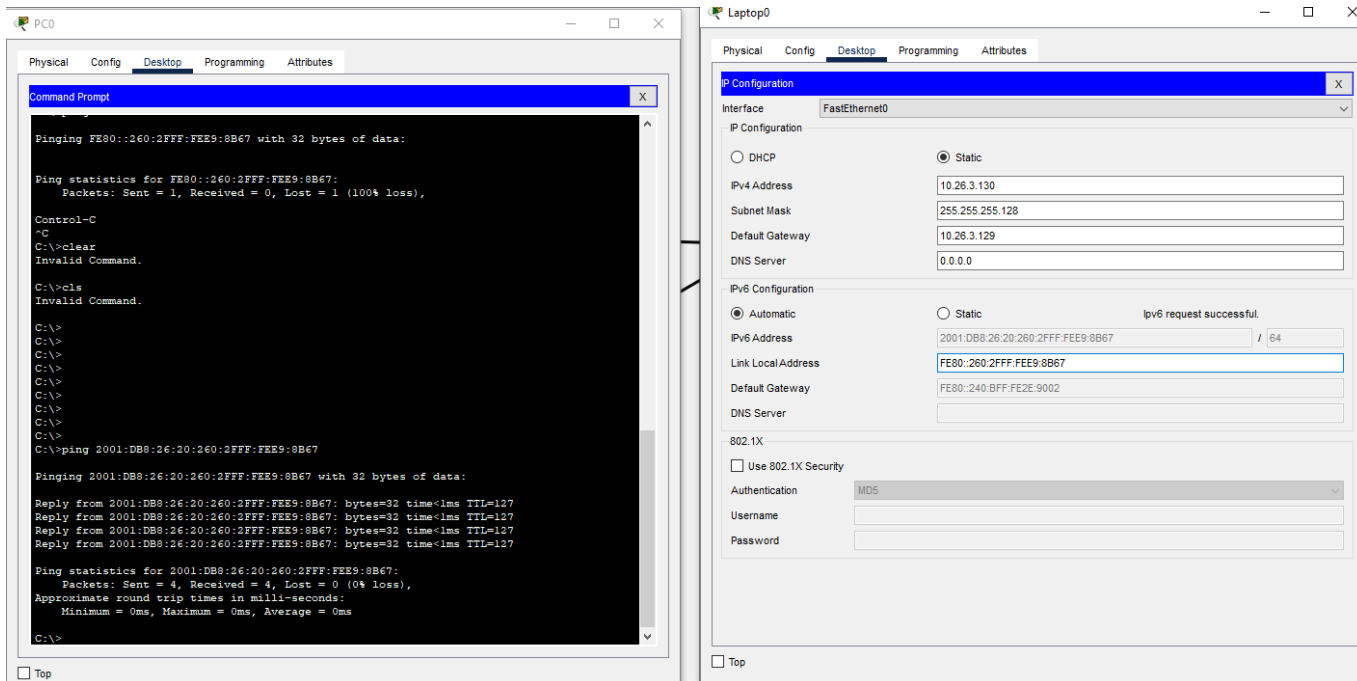
Switch#show ip dhcp pool v120
Pool v120 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 126
Leased addresses : 0
Excluded addresses : 2
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.26.3.129 10.26.3.129 - 10.26.3.254 0 / 2 / 126
Switch#show ip dhcp pool v110
Pool v110 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 126
Leased addresses : 2
Excluded addresses : 2
Pending event : none

1 subnet is currently in the pool
Current index IP address range Leased/Excluded/Total
10.26.3.1 10.26.3.1 - 10.26.3.126 2 / 2 / 126
Switch#

```

f. Prüft auch die IPv6-Connectivity!



Teil 2 – Übertrag das Szenario auf einen passenden L3-Switch im Labor.

1. Dokumentiert die notwendigen Konfigurationsschritte und dokumentiert auch deren Erfolg analog zum letzten Punkt der obigen PT-Anleitung.

```

Switch(config)#vlan 10
Switch(config-vlan)#name VLAN10
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name VLAN20
Switch(config-vlan)#exit
Switch(config)#interface gi1/0/1
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface gi1/0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 20
Switch(config-if)#
*Mar 1 00:04:49.985: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, cha
nged state to down
Switch(config-if)#no shutdown
Switch(config-if)#exit

Switch(config)#interface vlan 10
Switch(config-if)#
*Mar 1 00:05:08.675: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan10, ch
anged state to up ip address 10.26.3.1 255.255.255.128
Switch(config-if)#ip address 10.26.3.1 255.255.255.128
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface vlan 20
Switch(config-if)#
*Mar 1 00:05:47.800: %LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan20, ch
anged state to up
Switch(config-if)#ip address 10.26.3.129 255.255.255.128
Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#interface range gi1/0/1-4
Switch(config-if-range)#spanning-tree portfast

```

```
Switch(config)#ip routing
Switch(config)#ip dhcp pool v110
Switch(dhcp-config)#network 10.26.3.0 255.255.255.128
Switch(dhcp-config)#default-router 10.26.3.1
Switch(dhcp-config)#dns-server 10.10.80.1
Switch(dhcp-config)#dns-server 10.10.80.2
Switch(dhcp-config)#exit
Switch(config)#ip dhcp pool v120
Switch(dhcp-config)#network 10.26.3.128 255.255.255.128
Switch(dhcp-config)#default-router 10.26.3.129
Switch(dhcp-config)#dns-server 10.10.80.1
Switch(dhcp-config)#dns-server 10.10.80.2
Switch(dhcp-config)#exit
```

```
edv@nw-labor-02:~$ ping 10.26.3.130
PING 10.26.3.130 (10.26.3.130) 56(84) bytes of data.
64 bytes from 10.26.3.130: icmp_seq=1 ttl=63 time=0.272 ms
64 bytes from 10.26.3.130: icmp_seq=2 ttl=63 time=0.367 ms
64 bytes from 10.26.3.130: icmp_seq=3 ttl=63 time=0.424 ms
^C
--- 10.26.3.130 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2039ms
rtt min/avg/max/mdev = 0.272/0.354/0.424/0.062 ms
edv@nw-labor-02:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: eno1: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 98:90:96:b2:96:b7 brd ff:ff:ff:ff:ff:ff
    altname enp0s25
    inet 10.26.3.2/25 brd 10.26.3.127 scope global dynamic noprefixroute eno1
        valid_lft 86181sec preferred_lft 86181sec
    inet6 fe80::3f8b:d1ee:2a23:6a62/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
edv@nw-labor-02:~$
```

```
edv@nw-labor-02:~$ ping 10.26.3.129
PING 10.26.3.129 (10.26.3.129) 56(84) bytes of data.
64 bytes from 10.26.3.129: icmp_seq=1 ttl=255 time=2.70 ms
64 bytes from 10.26.3.129: icmp_seq=2 ttl=255 time=2.19 ms
^C
--- 10.26.3.129 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1000ms
rtt min/avg/max/mdev = 2.191/2.446/2.701/0.255 ms
edv@nw-labor-02:~$ ping 10.26.3.1
PING 10.26.3.1 (10.26.3.1) 56(84) bytes of data.
64 bytes from 10.26.3.1: icmp_seq=1 ttl=255 time=2.02 ms
64 bytes from 10.26.3.1: icmp_seq=2 ttl=255 time=2.11 ms
64 bytes from 10.26.3.1: icmp_seq=3 ttl=255 time=0.664 ms
^C
--- 10.26.3.1 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2000ms
rtt min/avg/max/mdev = 0.664/1.597/2.107/0.660 ms
```

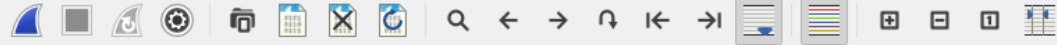
2. Zeichnet mit Wireshark eine IP-Initialisierung mit DHCP auf.

Dokumentiert die Phasen der IP-Adressvergabe:

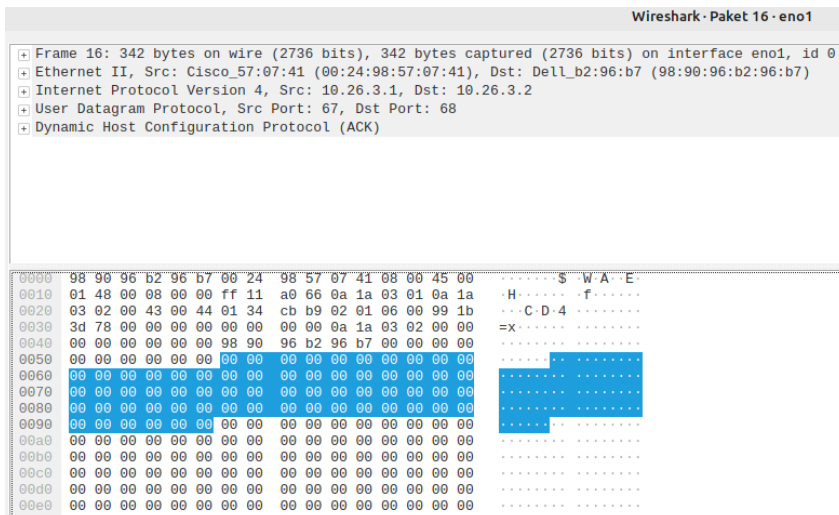
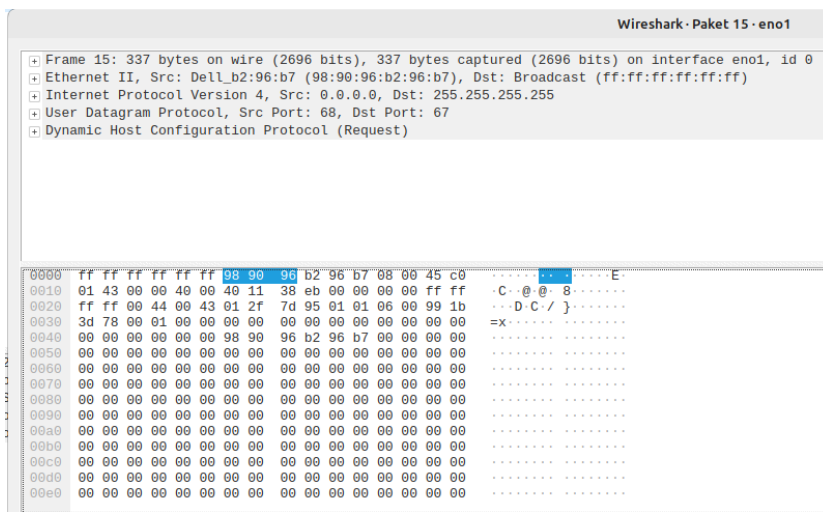
Discover / Offer / Request / Acknowledge

und untersucht die Pakete im Detail!

Da ich die adresse renewe habe ich nur request und ackn



No.	Time	Source	Destination	Proto	Length	Info
15	7.873942351	0.0.0.0	255.255.255.255	DHCP	337	DHCP Request - Transaction ID 0x991b3d78
16	7.875537980	10.26.3.1	10.26.3.2	DHCP	342	DHCP ACK - Transaction ID 0x991b3d78



- Hängt den PC an das Schulnetz (A-Dose) und untersucht welche weiteren Informationen via DHCP an den PC geliefert werden.

dhcp							
No.	Time	Source	Destination	Protocol	Length	Info	
9	8.772297820	0.0.0.0	255.255.255.255	DHCP	337	DHCP Request	- Transaction ID 0x8bb2d680
10	8.775855084	10.10.72.254	255.255.255.255	DHCP	342	DHCP NAK	- Transaction ID 0x8bb2d680
11	8.776993510	0.0.0.0	255.255.255.255	DHCP	337	DHCP Discover	- Transaction ID 0xd315ed1e
14	9.322342679	10.10.72.254	10.10.72.100	DHCP	353	DHCP Offer	- Transaction ID 0xd315ed1e
15	9.322498293	0.0.0.0	255.255.255.255	DHCP	343	DHCP Request	- Transaction ID 0xd315ed1e
16	9.328763958	10.10.72.254	10.10.72.100	DHCP	353	DHCP ACK	- Transaction ID 0xd315ed1e

Im Ack:

```

+ Frame 16: 353 bytes on wire (2824 bits), 353 bytes captured (2824 bits) on interface eno1, id 0
+ Ethernet II, Src: Cisco_ff:fd:8c (00:08:e3:ff:fd:8c), Dst: Dell_b2:96:b7 (98:90:96:b2:96:b7)
+ Internet Protocol Version 4, Src: 10.10.72.254, Dst: 10.10.72.100
+ User Datagram Protocol, Src Port: 67, Dst Port: 68
- Dynamic Host Configuration Protocol (ACK)
  Message type: Boot Reply (2)
  Hardware type: Ethernet (0x01)
  Hardware address length: 6
  Hops: 0
  Transaction ID: 0xd315ed1e
  Seconds elapsed: 0
+ Bootp flags: 0x0000 (Unicast)
  Client IP address: 0.0.0.0
  Your (client) IP address: 10.10.72.100
  Next server IP address: 10.10.7.10
  Relay agent IP address: 10.10.72.254
  Client MAC address: Dell_b2:96:b7 (98:90:96:b2:96:b7)
  Client hardware address padding: 000000000000000000000000
  Server host name not given
  Boot file name: undionly.kpxe
  Magic cookie: DHCP
+ Option: (53) DHCP Message Type (ACK)
- Option: (58) Renewal Time Value
  Length: 4
  Renewal Time Value: (1800s) 30 minutes
- Option: (59) Rebinding Time Value
  Length: 4
  Rebinding Time Value: (3150s) 52 minutes, 30 seconds
- Option: (51) IP Address Lease Time
  Length: 4
  IP Address Lease Time: (3600s) 1 hour
+ Option: (54) DHCP Server Identifier (10.10.80.1)
+ Option: (1) Subnet Mask (255.255.255.0)
- Option: (6) Domain Name Server
  Length: 8
  Domain Name Server: 10.10.80.1
  Domain Name Server: 10.10.80.2
- Option: (15) Domain Name
  Length: 13
  Domain Name: htlinn.local
- Option: (3) Router
  Length: 4
  Router: 10.10.72.254
- Option: (42) Network Time Protocol Servers
  Length: 4
  Network Time Protocol Server: 10.10.99.254
+ Option: (255) End

```

That's it, Bravo!

Bitte als NWL7-*Name*.PDF im moodle abgeben!